Digital logic

Higher · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ complete truth tables for AND, OR and NOT and combinations of gates
- ☐ turn a written problem into a logic circuit
- ☐ write a Boolean expression from a circuit or truth table
- ☐ use NAND and NOR equivalents to reduce the number of chips

I DO — Worked example

Read how a full-mark answer is built

A machine guard system: the motor runs (Z) only when the guard is closed (A) AND the start button is pressed (B), OR when the maintenance key (C) is on.

1 Draw the logic diagram. (3 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ AND gate with inputs A and B
- ✓ OR gate with the AND output and C
- ✓ Output Z from the OR gate

WE DO — Guided practice

Use the starters and word bank to help you

A circuit has inputs A and B into an OR gate whose output goes through a NOT gate to Z.

2 Complete the truth table for Z. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

NOR

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) How many rows does a 3-input truth table have?
☐ 3 ☐ 6 ☐ 8 ☐ 9

b) A=1, B=0 into an OR gate gives...
☐ 0 ☐ 1 ☐ Undefined ☐ 2

c) A NAND gate with its two inputs joined acts as a...
☐ NOT gate ☐ OR gate ☐ AND gate ☐ NOR gate

d) Which gate gives 1 only when its inputs differ?
☐ NAND ☐ NOR ☐ EOR ☐ AND

4 Key terms: write the letter of the matching description next to each term.

1. **AND gate**	____	**A** Output is the opposite of the input.
2. **OR gate**	____	**B** 8
3. **NOT gate**	____	**C** Output 1 only when all inputs are 0.
4. **Rows in a 3-input truth table**	____	**D** Output 1 only when all inputs are 1.
5. **NAND**	____	**E** Output 0 only when all inputs are 1.
6. **NOR**	____	**F** Output 1 when any input is 1.

A smoke effect turns on (Z = 1) when the stage door is closed (A = 1) and either the switch (B) or the timer (C) is on.

5 Write the Boolean equation for Z. (2 marks)

A circuit: Z = (A AND NOT B) OR C.

6 Write the Boolean expression and draw the NAND equivalent with the fewest gates. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?